

OpenClaw Servo Control Tutorial

Before starting, please complete the following in the `Pre-use Preparation` directory: [01-Peripheral Hardware Configuration], [02-OpenClaw API-KEY Configuration], [03-OpenClaw Switching Models], [04-AI Voice Interaction Configuration], [05-Three Chat Modes Configuration Testing]. If you are currently only using TUI / WhatsApp, you can skip `04` for now. This document assumes the basic environment is already prepared and only covers servo-related content.

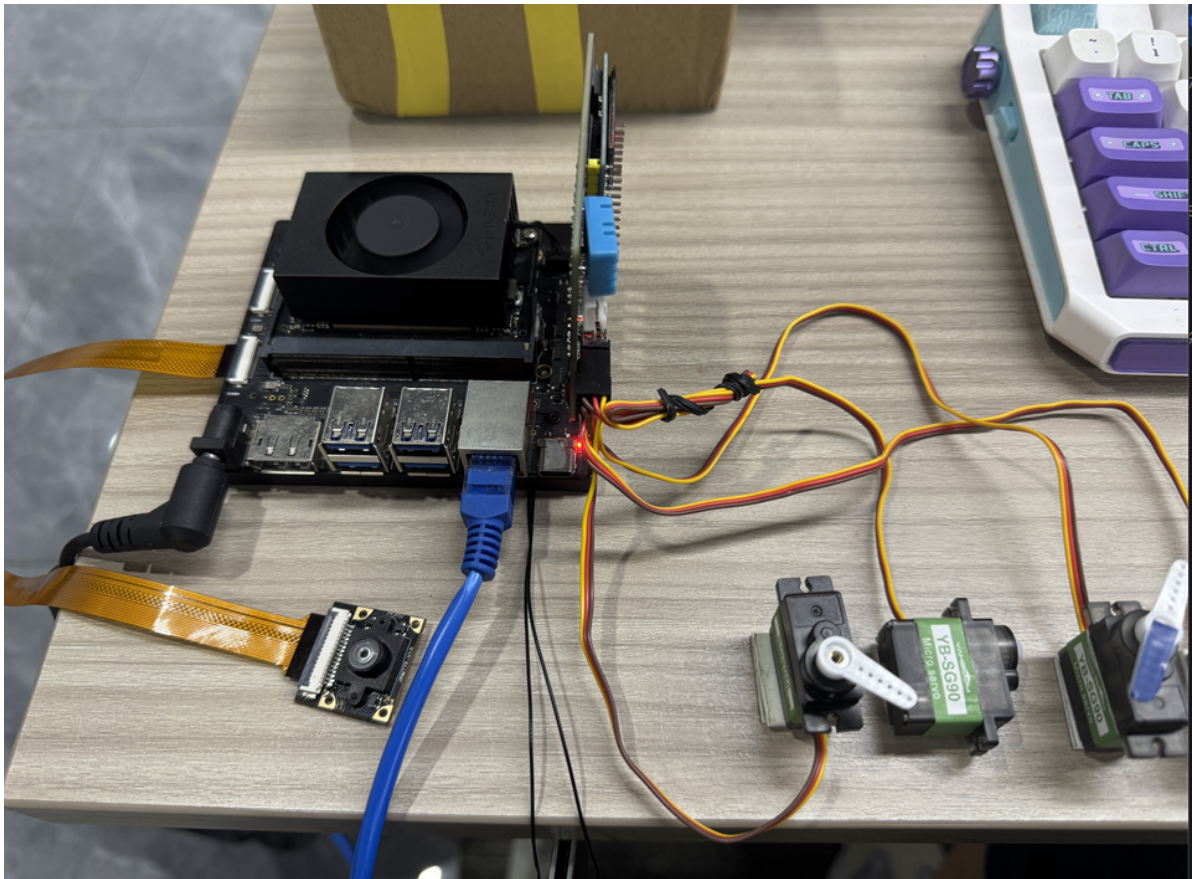
1. Tutorial Objectives

This tutorial demonstrates OpenClaw's natural language control capability through servo peripherals. After completing this tutorial, readers will be able to:

- Understand the basic approach of OpenClaw controlling servos
- Chat with OpenClaw through three methods: WhatsApp, TUI, and voice interaction
- Complete four servo examples: single servo control, servo dancing, yes/no Q&A, simulated table tennis, simulated pointer

2. Hardware Preparation

- OpenClaw main control board
- Servo `[1 / 4 units]`
- `[Optional]` Microphone module
- Wiring diagram



3. Software Preparation

Code entry points used in this tutorial

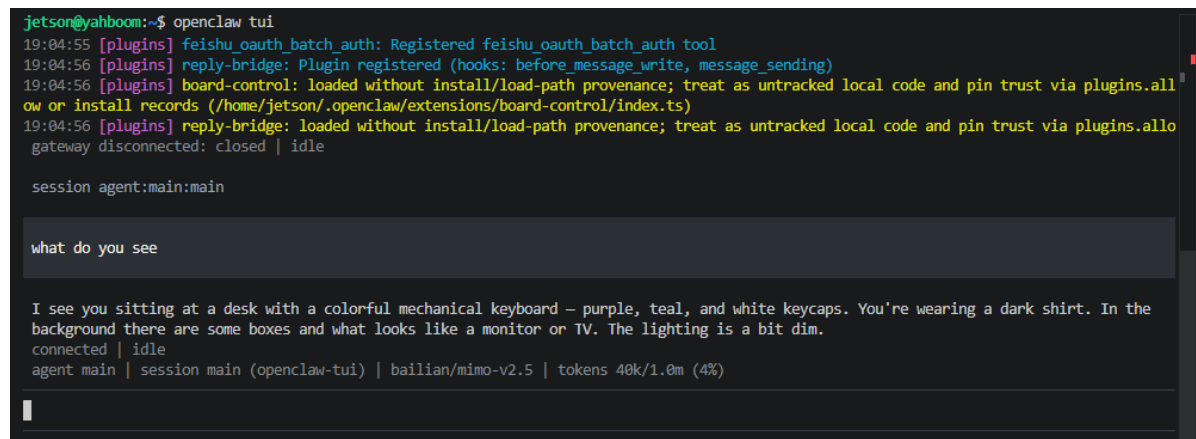
- Low-level driver code: `/home/jetson/hardware_hal_demo/`
- Python driver library: `/home/jetson/Downloads/python/`
- Voice entry: `/home/jetson/voice_to_openclaw/src/main.py`
- Terminal text entry: `/home/jetson/voice_to_openclaw/src/text_main.py`

4. Testing the Servo

Before starting this tutorial, please complete a basic `openclaw tui` chat self-check; if you haven't done so, see [05-Three Chat Modes Configuration Testing] first. After confirming OpenClaw can reply normally, proceed to the servo-specific test.

Enter the following command in the terminal to enter TUI:

```
openclaw tui
```



```
jetson@yahboom:~$ openclaw tui
19:04:55 [plugins] feishu_oauth_batch_auth: Registered feishu_oauth_batch_auth tool
19:04:56 [plugins] reply-bridge: Plugin registered (hooks: before_message_write, message_sending)
19:04:56 [plugins] board-control: loaded without install/load-path provenance; treat as untracked local code and pin trust via plugins.all
or install records (/home/jetson/.openclaw/extensions/board-control/index.ts)
19:04:56 [plugins] reply-bridge: loaded without install/load-path provenance; treat as untracked local code and pin trust via plugins.all
gateway disconnected: closed | idle

session agent:main:main

what do you see

I see you sitting at a desk with a colorful mechanical keyboard – purple, teal, and white keycaps. You're wearing a dark shirt. In the
background there are some boxes and what looks like a monitor or TV. The lighting is a bit dim.
connected | idle
agent main | session main (openclaw-tui) | baillian/mimo-v2.5 | tokens 40k/1.0m (4%)
```

After ensuring the hardware is connected, you can test servo control by entering the following in the terminal:

- Turn servo 1 to 0 degrees
- Turn the servo to 90 degrees
- Turn the servo to 180 degrees

Note: If possible, specify which servo number to control

```
File Edit Tabs Help

Rotate Servo 1 to 0 degrees.

Servo 1 rotated to 0 degrees. ✓

Rotate the servo to 180 degrees.

Servo 1 rotated to 180 degrees. ✓

Rotate all servos to 180 degrees.

All four servos rotated to 180 degrees simultaneously. ✓
connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens
43k/1.0m (4%)
```

If you can see the servo driving normally, it means all of the following are working properly:

- OpenClaw can receive commands
- Servo power supply is normal
- Servo angle control range is correct
- Clear feedback is visible after command execution

5. Three Methods to Chat with OpenClaw

Method	Suitable Scenarios	Advantages	Recommended Positioning
WhatsApp	Remote control, classroom demo	Easy to connect, suitable for multi-person experience	Demonstrate "remote natural language control"
TUI	Local debugging, command verification	Most direct, easiest to troubleshoot	First get the control pipeline working
Voice Code	Voice interaction, complete AI experience	Closest to real conversation experience	As the final demonstration method

If the AI voice module + speaker is an optional add-on, Feishu chat, TUI, and WhatsApp chat need to play back OpenClaw's replies. You can run the following program in the terminal. AI voice module chat does not need to run it again.

- Modify the broadcast parameter

```
/home/jetson/voice_to_openclaw/.env
```

```
ENABLE_TTS=true # This parameter controls whether to broadcast. After
modification, save and run the following program in the terminal
```

- Enter in terminal

```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```

```
• (.venv) jetson@yahboom:~/voice_to_openclaw$ python -m src.openclaw_session_tts
[系统] OpenClaw 会话播报器已启动
[系统] 监听目录=/home/jetson/.openclaw/agents/main/sessions
[系统] 监听渠道=feishu,tui,whatsapp
```

5.1 whatsapp Method

For WhatsApp connection, QR code creation, and gateway restart steps, please refer to [05-Three Chat Modes Configuration Testing]. This section only retains chat examples suitable for the servo tutorial.

You can chat directly, for example:

- Turn servo 1 to 0 degrees
- Turn the servo to 90 degrees
- Make the servo swing left and right

19:42



< 2



+86 153 3885 7526 (你)

给自己发消息

[error-code#overdue-payment](#)

19:40 ✓✓

Make the servo swing left and right .

19:41 ✓✓

The sweep action isn't available. I'll create the swinging motion using multiple angle changes with delays.

19:42 ✓✓

Done! The servo completed a left-to-right swing:

- Moved to 0° (left)
- Swung to 180° (right)
- Returned to 90° (center)

The servo is now ready for the next command.

19:42 ✓✓



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5.2 TUI Method

If you have completed the TUI self-check in Part 0, simply enter the following command in the terminal to continue chatting:

```
openclaw tui
```

```
sunrise@ubuntu:~/openclaw$ openclaw tui
🔌 OpenClaw 2026.3.24 (cff6dc9) – You had me at 'openclaw gateway start.'
openclaw tui - ws://127.0.0.1:18789 - agent main - session main
session agent:main:main

hello

Hello, master! ☺ I'm Xiao Ya, ready to serve.
is there anything I can help you with?
gateway connected | idle
agent main | session main (openclaw-tui) | deepseek/deepseek-v4-pro | tokens 12k/1.0m (1%)
```

5.3 Voice Code Method

Requires optional AI voice interaction module + speaker

For voice module installation, environment configuration, and basic wake process, please refer directly to [04-AI Voice Interaction Configuration]. Voice mode is more suitable for demonstrating spoken servo commands, for example:

- Turn the servo to 90 degrees
- Make the servo nod
- Make the servo swing left and right

```
2026-05-14 14:48:20 [INFO] voice_to_openclaw:run:26 - voice_to_openclaw startin
[系统] voice_to_openclaw 已启动
[系统] ASR=xunfei_ws/iat/en_us | 发布=gateway_chat | TTS=开/xunfei_ws | 唤醒=串口
[系统] 串口唤醒已启用: /dev/ttyUSB0@115200
[提示] 说“清空上下文”可以重置当前会话
[唤醒] 检测到串口信号, 开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 6060 ms 语音, 正在转写

[你] Make the servo swing left and right.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] Let me make the servo swing using individual angle commands:
Done!
The servo swung left to right and back to center. 🌀

The motion
sequence was:
- Center (90°) → Left (0°) → Right (180°) → Center
(90°)
[信息] 回复已裁剪为播报文本: 202 -> 116 字
```

6. Comprehensive Cases

Note: The following are all demonstrated using `openclaw_tui` for chat (you can choose your preferred chat method)

6.1 Case 1: Servo Dancing

Experimental Objective

Make the servo move continuously according to a set of preset angles, creating a rhythmic "dancing" effect.

Effect Description

The servo can switch between multiple angles, such as swinging left, centering, swinging right, rapid shaking, and then returning to the initial position. Even with just one servo, the changes in movement rhythm can still create a noticeable "dancing feel".

Example Commands

- Make the servo dance for 10 seconds
- Make the servo swing left and right, with a faster rhythm
- Arrange a three-beat dance step for the servo

6.2 Case 2: Yes/No Q&A

Experimental Objective

Make the servo express "yes" or "no" with movements, such as nodding for "yes" and shaking for "no".

Effect Description

When OpenClaw determines the answer is affirmative, the servo performs a nodding action; when the answer is negative, the servo performs a head-shaking action. This case is very suitable for highlighting the combination of "natural language understanding + action expression".

Example Commands

- If the answer is "yes", make the servo nod
- If the answer is "no", make the servo shake its head
- Help me judge this question and answer with servo movements

6.3 Case 3: Simulated Table Tennis

Experimental Objective

Make the servo continuously move back and forth, simulating the feeling of a ping pong ball bouncing back and forth across the table.

Effect Description

The servo oscillates at a constant or variable speed between the left and right boundary angles. If accompanied by voice descriptions or screen prompts, the audience will easily understand this as a "ping pong ball back and forth" simulation.

Example Commands

- Make the servo simulate a ping pong ball bouncing back and forth
- Swing left and right 20 times, with a faster speed
- Start from the left, do a back and forth movement demo

6.4 Case 4: Simulated Pointer

Experimental Objective

Use the servo as a rotatable pointer to indicate values, directions, or states.

Effect Description

This case can very intuitively demonstrate the mapping from "abstract values -> actual angles". For example, mapping values from 0 to 100 to 0 to 180 degrees, the servo can point to the target position like a gauge needle.

Example Commands

- Turn the pointer to 30%
- Make the servo point to the 90 degree position
- Simulate a dashboard, the current value is 75